

Find Eigen vector and value of

$$1. A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\lambda = ?$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Now, } A = \lambda I$$

$$\text{or } |A| = |\lambda I| = 0$$

$$\text{or, } \left| \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\text{or, } \left| \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\text{or, } \begin{vmatrix} 4-\lambda & 1-0 \\ 2-0 & 3-\lambda \end{vmatrix} = 0$$

$$\text{or } \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\text{or, } (4-\lambda)(3-\lambda) - 2 = 0$$

$$\text{or, } 12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\text{or, } 10 - 7\lambda + \lambda^2 = 0$$

$$\text{or } \lambda^2 - 7\lambda + 10 = 0$$

$$\text{or, } (\lambda - 5)(\lambda - 2) = 0$$

So, Eigen values are

$$\lambda = 5$$

$$\lambda = 2$$

$$V_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_2 = 1, x_1 = 1$$

For Eigen Vector

$$\text{For, } \lambda = 5.$$

$$AV - \lambda V = 0$$

$$\text{or } (A - \lambda I)V = 0$$

$$\text{or, } (A - 5I)V = 0$$

$$\text{let } v = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\text{or, } \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\text{or, } \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\text{or, } \begin{bmatrix} 4-5 & 1-0 \\ 2-0 & 3-5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\text{or, } \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\text{or, } (-x_1 + x_2)(2x_1 - 2x_2) = 0$$

Now, or,

$$-x_1 + x_2 = 0$$

$$\text{let } x_1 = 1$$

$$-1 + x_2 = 0$$

$$x_2 = 1$$

$$2x_1 - 2x_2 = 0$$

$$2x_1 - 2x_2 = 0$$

$$2 = 2x_2$$

$$2 = x_2 \text{ or, } 1 = x_2$$

for $\lambda = 2$

$$A - 2I$$

$$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\text{or } \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 0$$

$$\text{or } \begin{bmatrix} 4-2 & 1-0 \\ 2-0 & 3-2 \end{bmatrix} = 0$$

$$\text{or } \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} = 0$$

$$\text{Now, } (2x_1 - 2x_2) = 0$$

$$\text{Let } x_2 = 1$$

$$\text{Let } x_1 = 1$$

$$2x_1 - 2 \times 2 = 0$$

$$2 \times 1 - 2x_2 = 0$$

$$\text{or } 2x_1 - 4 = 0$$

$$\text{or } 2 - 2x_2 = 0$$

$$\text{or } 2x_1 = 4$$

$$\text{or } 2 = 2x_2$$

$$\text{or } x_1 = \frac{4}{2}$$

$$\text{or } \frac{2}{2} = x_2$$

$$\text{or } x_1 = 2$$

$$\text{or } 1 = x_2$$

$$\therefore x_1 = 2, x_2 = 1, V = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$2. \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

$$\lambda = ?$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Now } m = \lambda I$$

$$\text{or } |m| = |\lambda I| = 0$$

$$\text{or } \left| \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\text{or } \left| \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\text{or } \begin{vmatrix} 1-\lambda & 2-0 \\ 3-0 & 2-\lambda \end{vmatrix} = 0$$

$$\text{or } (1-\lambda)(2-\lambda) - 6 = 0$$

$$\text{or } 2 - \lambda - 2\lambda + \lambda^2 - 6 = 0$$

$$\text{or } \lambda^2 - 3\lambda - 4 = 0$$

$$\text{or } (\lambda - 4)(\lambda + 1) = 0$$

So, Eigen values are

$$\lambda_1 = 4$$

$$\lambda_2 = -1$$

Now, For Eigen Vectors

$$\lambda = 4$$

$$(m - 4I)v = 0, \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\text{or } \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} v = 0 \Rightarrow \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} v = 0$$

$$\text{let } v = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{or, } \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\text{or, } (-3x + 2y)(3x - 2y) = 0$$

Therefore,

$$-3x + 2y = 0$$

$$\text{or, } 2y = 3x$$

$$\text{or, } y = \frac{3}{2}x$$

$$\therefore y = 3$$

$$x = 2$$

$$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\text{For } \lambda = -1$$

$$(M - (-1)I)v = 0$$

$$\text{or, } (M + I)v = 0$$

$$\text{or, } \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} v = 0$$

$$\text{or, } \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\text{or, } (2x + 2y)(3x + 3y) = 0$$

Therefore,

$$2x + 2y = 0$$

$$\text{or, } 2x = -2y$$

$$\text{or, } \cancel{2x} = \cancel{-2y} \Rightarrow x = \frac{-2y}{-2}$$

$$\text{or, } x = y \quad \text{or, } x = 1, y = -1. \quad v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

5 $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ $\lambda = ?$ $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$M - \lambda I = 0$

or, $|M| = |\lambda I| = 0$

or, $\left| \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$

or, $\left| \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$

or, $\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$

or

$(2-\lambda)(2-\lambda) - 1 = 0$

or, $4 - 2\lambda - 2\lambda + \lambda^2 - 1 = 0$

or, $\lambda^2 - 4\lambda + 3 = 0$

or, $(\lambda - 3)(\lambda - 1) = 0$

$\therefore$ Eigen values are

$\lambda = 3$ $(\lambda - 3) = 0$ $\lambda = 3$

$\lambda_2 = 1$ $(\lambda - 1) = 0$ $\lambda = 1$

For Eigen Vector for $\lambda = 3$

$(M - 3I)v = 0$

or, $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} v = 0$

or, $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} v = 0$

or, $\begin{bmatrix} 2-3 & 1-0 \\ 1-0 & 2-3 \end{bmatrix} v = 0$, or, $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} v = 0$

Let $v = \begin{bmatrix} x \\ y \end{bmatrix}$

or, $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

or, $(x+y)(x-y) = 0$

So,

$-x+y = 0$

or, $y = x$

or, $x = 1$

or

$y = 1$

so, $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Eigen vector for $\lambda=1$

$$(M - 1I)v = 0$$

$$\text{or } (M - 1I)v = 0$$

$$\text{or } \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} v = 0$$

$$\text{or } \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} v = 0$$

$$\text{Let } v = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{or } \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\text{or } (x+y)(x+y) = 0$$

Therefore,

$$x+y = 0$$

$$\text{Let } x = 1$$

$$1+y = 0$$

$$\text{or } y = -1$$

$$\text{Then, } x+y = 0$$

$$\text{or } x-1 = 0$$

$$\text{or } x = 1$$

$$\therefore v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$